

MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 12: Friday 18 September 2026

(Week 6)

Remark. Q11.2 to Q11.4 are leftover exercises from the Tuesday class.

Remark. Q012.1 to Q12.2 are still on Section 13.8

Q11.2. Use the method of Lagrange multipliers to find the maximum value of the function $f(\mathbf{x}, y) = \mathbf{x}y$ subject to the constraints

$$(\mathbf{x} - 2)^2 + y^2 = 4 \quad \text{and} \quad \mathbf{x} \leq 2. \quad (*)$$

[Remark. The above question is modified from Q11.1 by adding an additional constraint $\mathbf{x} \leq 2$. So you may find your calculations in Q11.1 useful.]

Q11.3. Find the point(s) on the sphere $\mathbf{x}^2 + y^2 + \mathbf{z}^2 = 1$ furthest from the point $(2, 1, 2)$.

Q11.4. (Q08.5. revisited) A rectangular box with its top left open has its five faces made from different materials. The front face and the back face are made from materials costing 10 dollars per square meter. The two lateral faces are made from materials costing 5 dollars per square meter. The bottom face is made from materials costing 30 dollars per square meter. Using the method of Lagrange multipliers, find the minimum material cost of the rectangular box if the volume of the box is 6 cubic meters. Give also the dimensions (i.e. length, width, and height) of such box.

Q12.1. A rectangular box is placed in the first octant so that one corner **Q** is at the origin and the three sides adjacent to **Q** lie in the coordinate planes. The corner **P** diagonally opposite to **Q** lies on the ellipsoid

$$\frac{x^2}{3} + \frac{y^2}{9} + \frac{z^2}{4} = 1.$$

Use the method of Lagrange multipliers to determine the largest possible volume of the rectangular box.

Q12.2. Let $f(x, y) = x^2 - y^2$ be defined on the $D := \{(x, y) : x^2 + y^2 \leq 1\}$.

(a) Find the critical points of f .

(b) What are the maxima and minima of f on D ?

[Remark: Lagrange multipliers is used in part (b).]